

Backlog Buddy

Demonstration question paper generated from available previous papers - for practice only

B.Tech. I Semester Examination, 2025

Regulation: R20 | University: JNTUK

Data Structures (21CS32)

Time: 3 hours Max. Marks: 60

Answer one question from each unit. All questions carry equal marks.

UNIT - 1 [Introduction & Linear Structures]

- a) Explain the operations performed on a singly linked list with algorithms.
- b) Explain the operations performed on a singly linked list with algorithms.

UNIT - 2 [Stacks, Queues & Recursion]

- a) Write an algorithm to convert an infix expression into postfix form using a stack.
- b) Write an algorithm to convert an infix expression into postfix form using a stack.

UNIT - 3 [Trees & Balanced Search Trees]

- a) Explain AVL tree rotations with examples.
- b) Explain B-tree insertion and deletion with an example.

UNIT - 4 [Graphs & Graph Algorithms]

- a) Find the minimum spanning tree of the given graph using Kruskal algorithm.
- b) Explain Dijkstra algorithm for single source shortest path with an example.

UNIT - 5 [Searching, Sorting & Hashing]

- a) Write the quick sort algorithm and analyse its time complexity.
- b) Compare linear probing and quadratic probing used in hashing.